

《Automotive Configuration》

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Chapter 11 Control System of Harmful Emissions in Engine Exhaust

第11章 发动机有害排放物的控制系统

11.1 Harmful Emissions/pollutants in Automotive Engine Exhaust

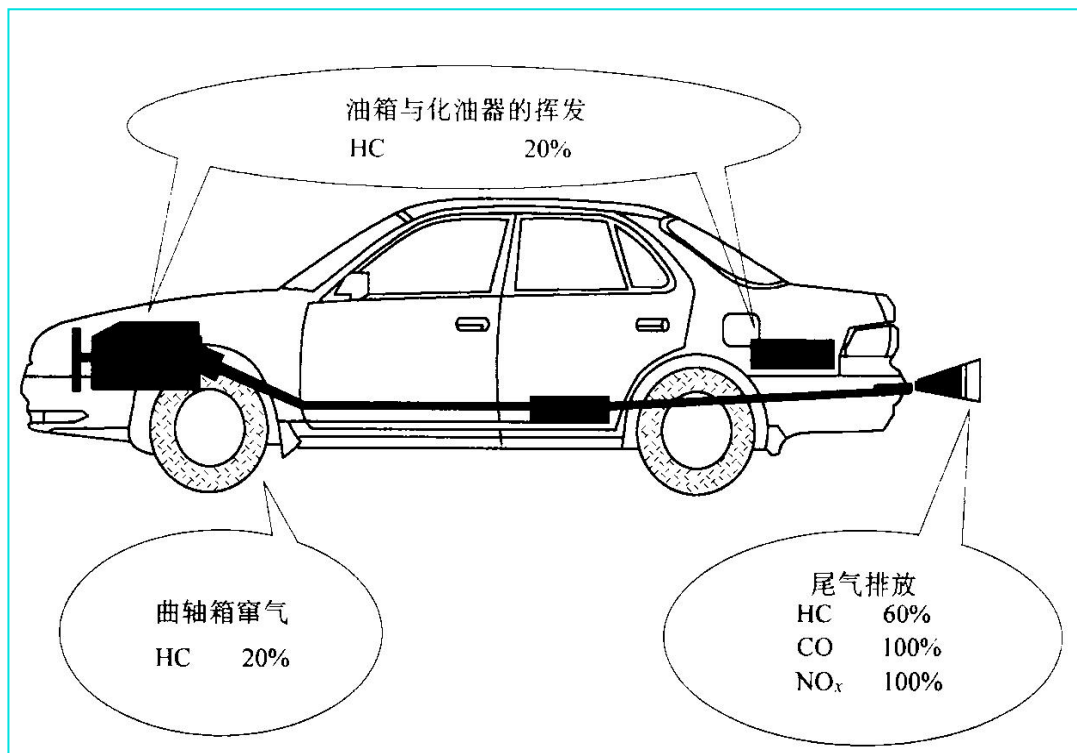
汽车发动机的有害排放物

11.2 Emissions Control Devices in Gasoline Engine

汽油机的排放控制装置

11.3 Emissions Control Devices in Diesel Engine

柴油机的排放控制装置



Tier	Date	CO	THC	NMHC	NOx	HC+NOx	PM	P _{max}
Diesel								
Euro 1†	July 1992	2.72 (3.16)	-	-	-	0.97 (1.13)	0.14 (0.18)	-
Euro 2	January 1996	1.0	-	-	-	0.7	0.08	-
Euro 3	January 2000	0.64	-	-	0.50	0.56	0.05	-
Euro 4	January 2005	0.50	-	-	0.25	0.30	0.025	-
Euro 5	September 2009	0.500	-	-	0.180	0.230	0.005	-
Euro 6 (future)	September 2014	0.500	-	-	0.080	0.170	0.005	-
Petrol (Gasoline)								
Euro 1†	July 1992	2.72 (3.16)	-	-	-	0.97 (1.13)	-	-
Euro 2	January 1996	2.2	-	-	-	0.5	-	-
Euro 3	January 2000	2.3	0.20	-	0.15	-	-	-
Euro 4	January 2005	1.0	0.10	-	0.08	-	-	-
Euro 5	September 2009	1.000	0.100	0.068	0.060	-	0.005**	-
Euro 6 (future)	September 2014	1.000	0.100	0.068	0.060	-	0.005**	-

European emission standards for passenger cars (g/km)

§ 11.1 Harmful Emissions in Automotive Engine Exhaust

汽车发动机的有害排放物

(1) HC

Include unburnt and partial-burnt **fuel** and vapor of lubricating **oil**.

包括未燃和未完全燃烧的燃油和润滑油蒸汽。

HC and NO_x can form photochemical smog(光化学烟雾) in sunlight, which is harmful to human health and environment.

HC formation of in-cylinder combustion :

lack of oxygen, crevice effect (缝隙效应), chilling and quenching (激冷与淬熄).

Source:

- ✓ HC in exhaust pipes accounts for $\approx 60\%$ of the total HC
- ✓ HC leakage from crankcase (曲轴箱窜气) accounts for $\approx 25\%$
- ✓ HC vapor from fuel supply system(供油系统的HC蒸汽) accounts for $\approx 15\% \sim 20\%$

(2) CO

Hydrocarbon fuel uncompleted combustion product, it will reduce the ability to absorb and transport oxygen in blood, after being inhaled(人体吸入后).

Formation
mechanism:
(形成机理)

- ✓ Oxygen deficit(缺氧);
- ✓ Low combustion temperature;
- ✓ Short combustion time;
- ✓ Inhomogeneous mixture(混合不均).

CO is mainly influenced by coefficient of excess air (过量空气系数): ϕ_a

(3) NO_x

Composition: NO (95%), NO_2 .

Formation mechanism:

- ◆ High temperature
- ◆ High oxygen concentration
- ◆ Long duration at high temperature & oxygen concentration

NO_x and HC can form photochemical smog(光化学烟雾) in sunlight, which is harmful to human health and environment.

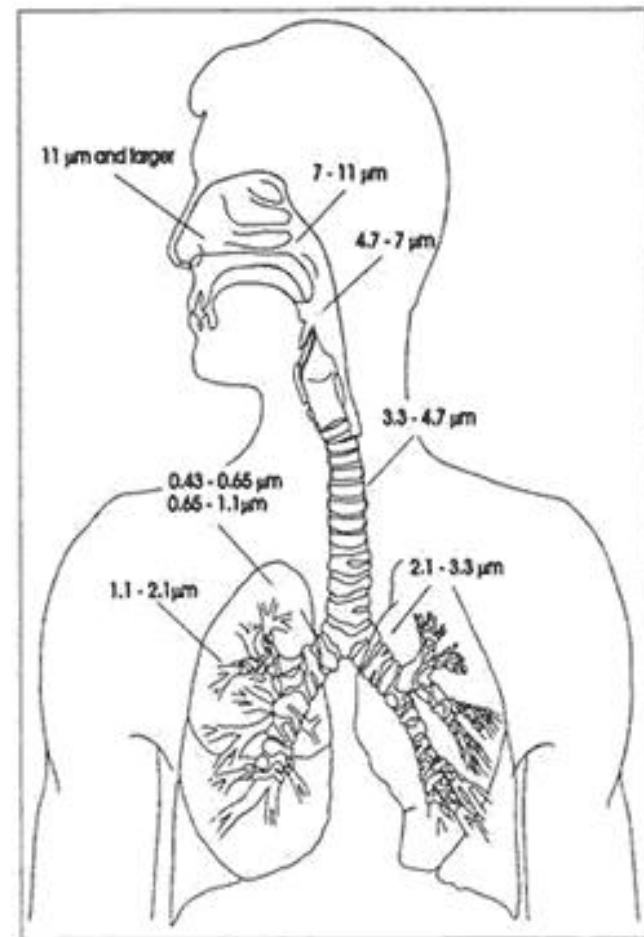
(4) Particulate Matter (PM)

Dry soot(碳烟) in diesel engine exhaust + soluble organic fraction (SOF,可溶性有机物) adsorbed on soot surface(表面吸附的可溶性有机物)

Harm: particles below $0.1\ \mu\text{m}$ are the most harmful to human body, after being inhaled into lung, they will be adsorbed on the lung cells.

SOF, polycyclic aromatic hydrocarbons(多环芳香烃) are cancer-causing toxins(致癌物质).

Formation mechanism: soot is cracked(裂解) by hydrocarbon fuel at high temperature and oxygen deficit (高温缺氧) conditions.



(5) SO_x (SO_2 , SO_3)

Harm:

- SO_x are harmful to human body.
- After being emitted into the air, they will form acid rain(酸雨), moreover, 50% of SO_x will become sulfates(硫酸盐) which is part of PM.
- SO_x will poison three-way catalyst(三效催化剂) in gasoline engine after-treatment system.

SO_x come from the sulphur (硫) in the fuel (Gasoline or diesel, lubricant) oxidated by oxygen in combustion process.

Sulfur content in the fuel (Gasoline or diesel, oil) has been reduced to ultralow level ($\approx 10\text{-}100\text{ppm}$).

Types of harmful emissions : CO、 HC、 NO_x、 PM、 SO_x.

Green-house emissions: CO₂, CH₄, ...

§ 11.2 Emissions Control Devices in Gasoline Engine

汽油机的排放控制装置

1. **Gasoline evaporation control system** (汽油蒸发控制系统);
2. **Positive Crankcase Ventilation system (PCV)**, 强制曲轴箱通风系统);
3. **Exhaust Gas Recirculation (EGR) System**(废气再循环系统);
4. **Three-way Catalytic converter**(TWC, 三效催化转化装置);
5. **Gasoline particulate filter (GPF)**, 汽油颗粒过滤器)

1 Gasoline Evaporation control system

(汽油蒸发控制系统)

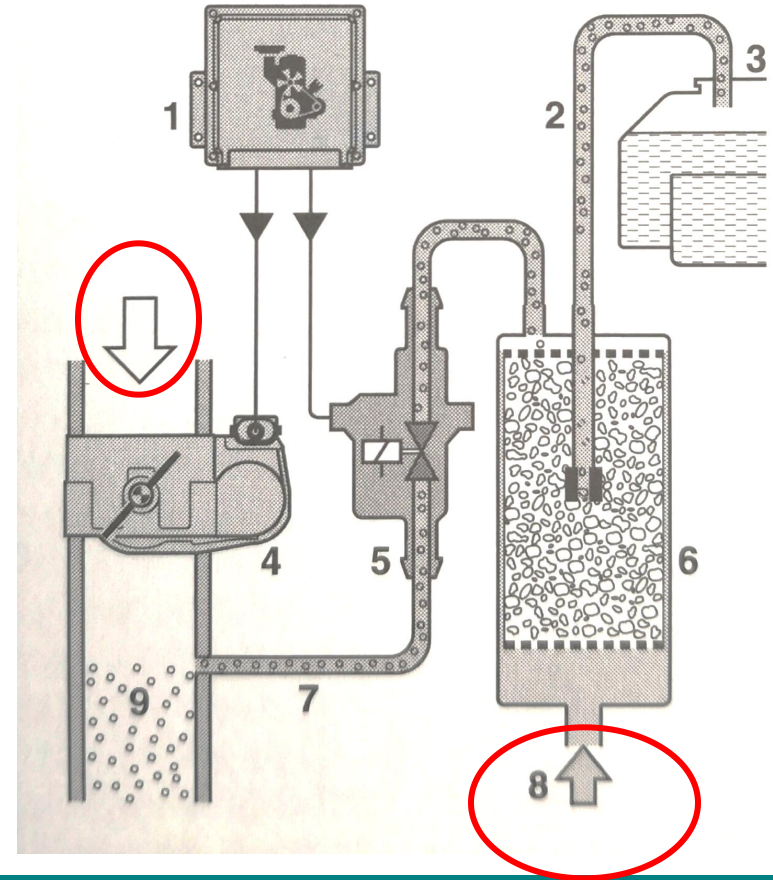
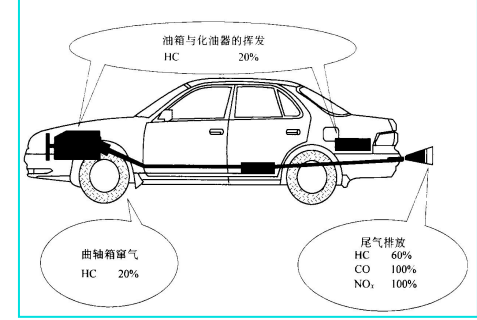
Function:

Prevent the fuel vapor from fueling system (fuel tank, etc.) diffusing into air.

Solution:

Collect the gasoline vapor and store it in the activated carbon canister(活性炭罐), then send it into cylinder and be burnt when the engine is running.

- 1 Engine ECU,
- 2 Fuel-tank vent line,
- 3 Fuel tank, 4 Throttle valve,
- 5 Purge valve, 6 Carbon canister,
- 7 Line to intake manifold,
- 8 Fresh air, 9 Intake manifold.

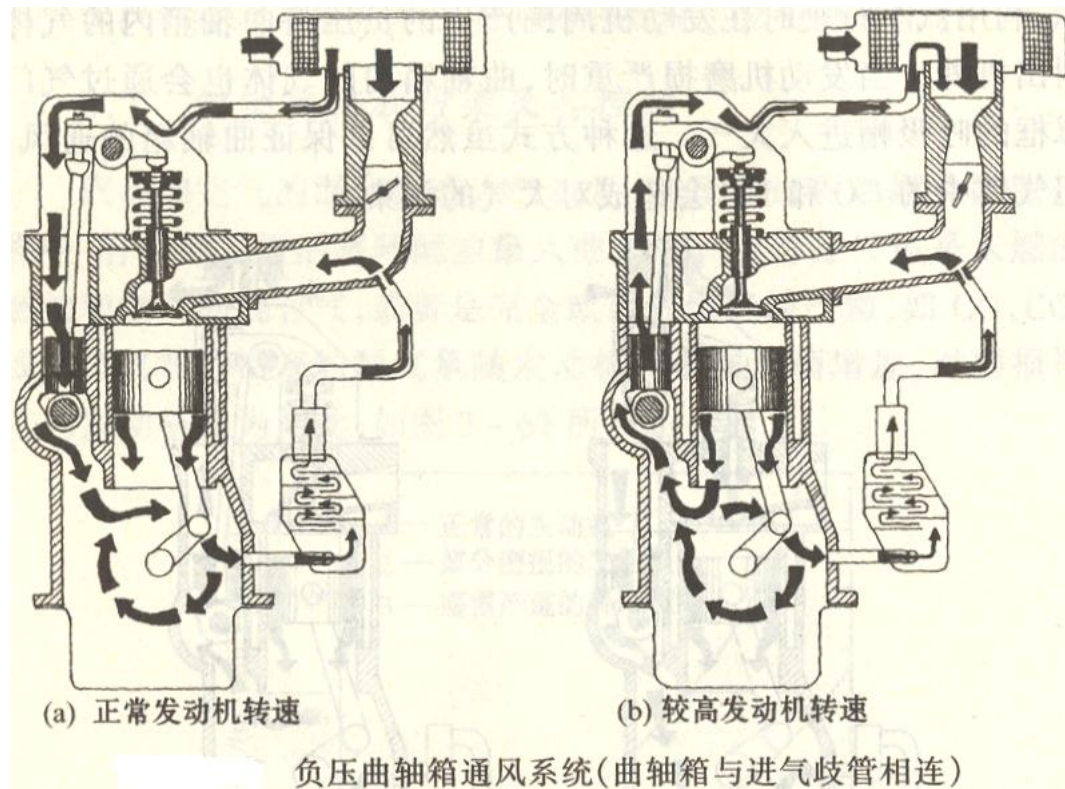


2 Positive Crankcase Ventilation system (PCV)

强制式曲轴箱通风装置（PCV系统）

Fundamental:

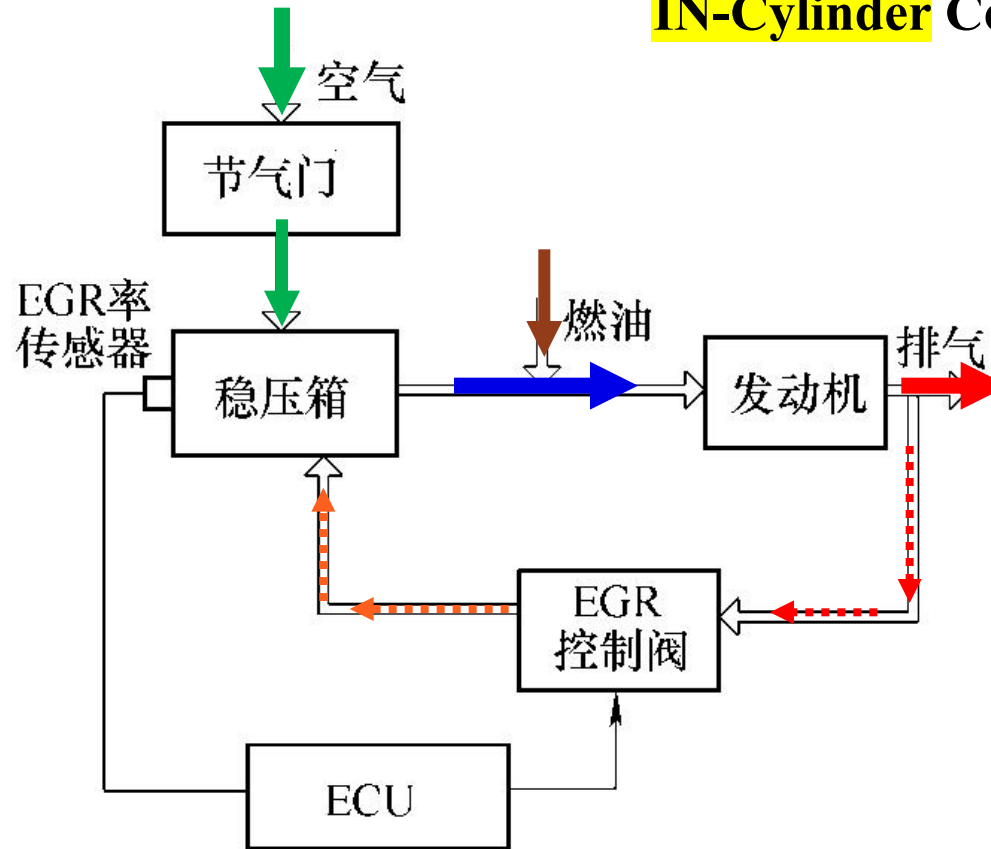
When the engine is running, send part of the fresh air from the air filter(空气滤清器) to cylinder head cover(气缸盖罩) by the hose(软管), then the air goes into the crankcase(曲轴箱) through the holes that are connecting cylinder head and body, and mix with crankcase gas, like lubricant vapor. Finally, the mixtures go into the intake manifold through PCV valve.



3 Exhaust Gas Recirculation System(废气再循环系统)

Basic function of EGR: $\downarrow \text{NO}_x$.

IN-Cylinder Control Method



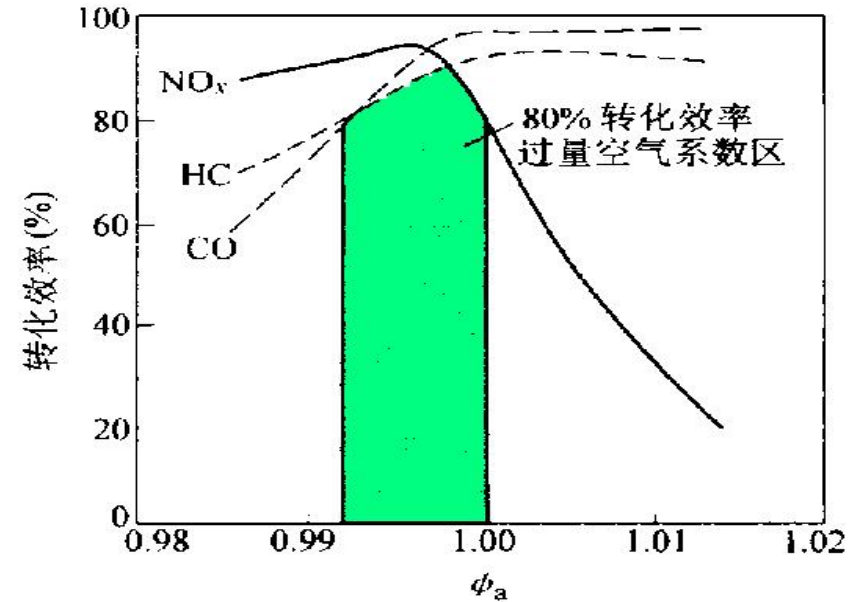
NOT popular in gasoline engines

4 Three-way catalytic converter (TWC, 三效催化转换装置)

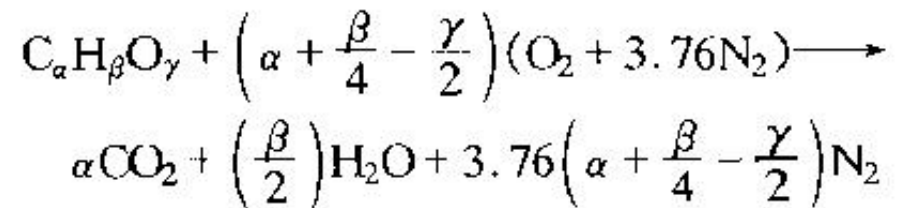
TWC can reduce CO, HC and NO_x emissions at the same time.

NO_x → N₂ + O₂, using CO+HC in the exhaust gas as the reductant (还原剂), while CO +HC are oxidized to CO₂ + H₂O.

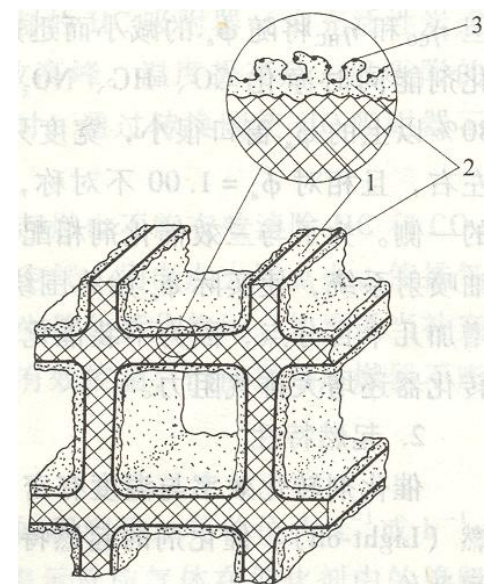
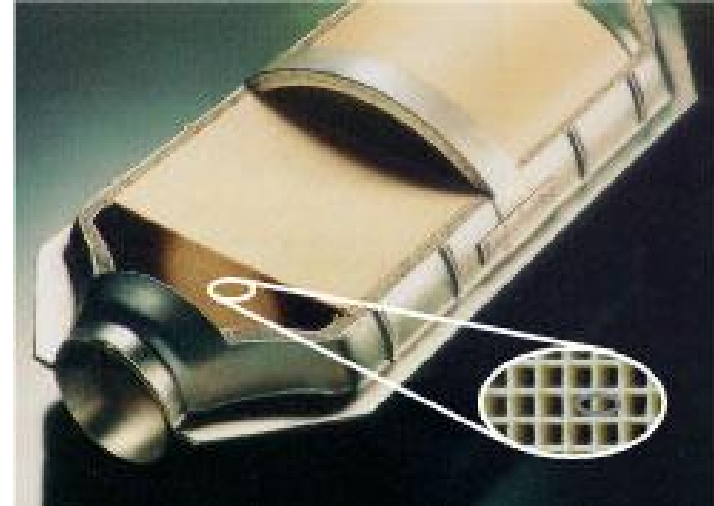
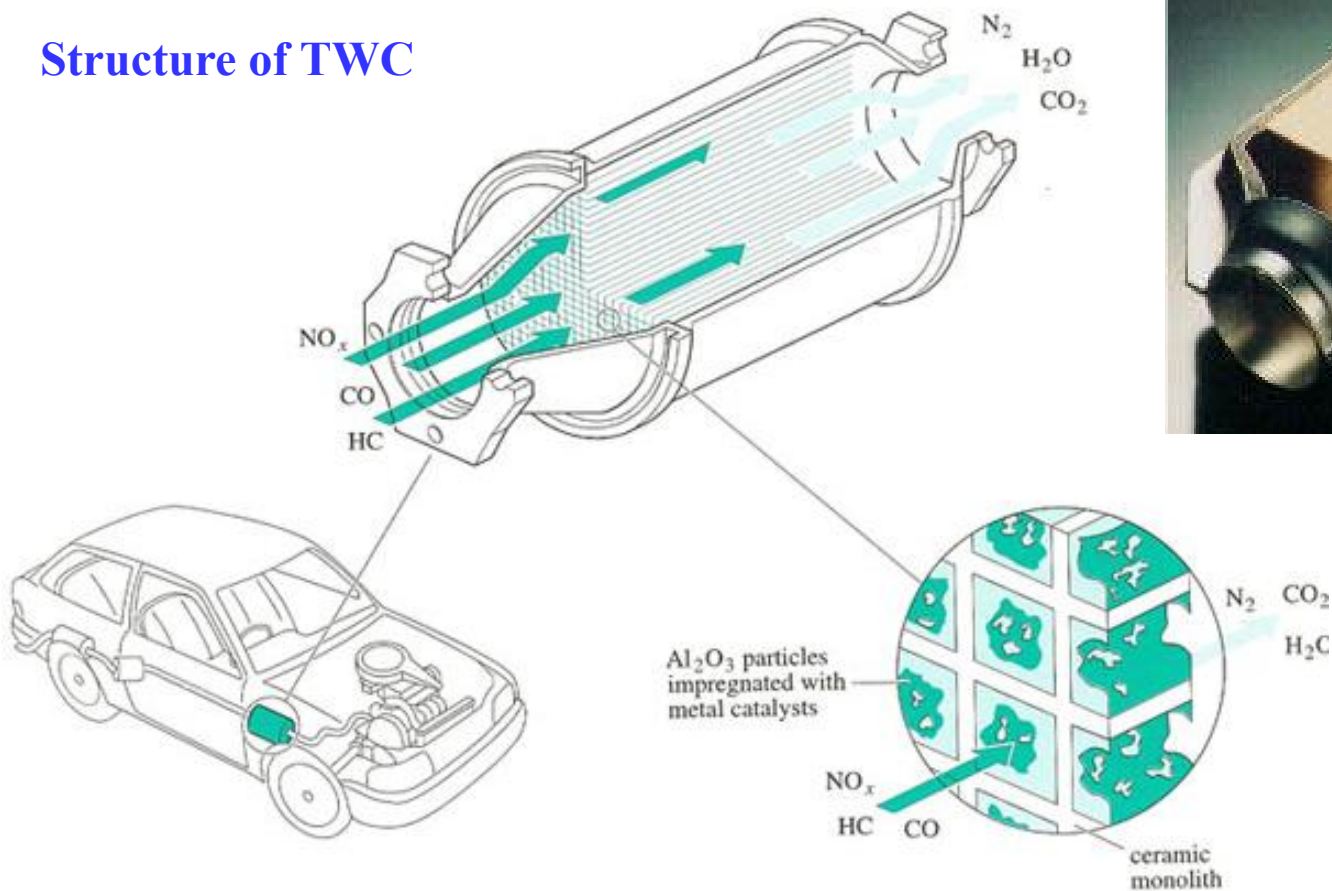
When air-fuel ratio (A/F) = 14.7 for gasoline engines, the oxidation - reduction reaction reaches equilibrium, CO, HC and NO_x transform efficiencies reach the highest level at the same time.



Exhaust gas conversion efficiency and ϕ_a



Structure of TWC



外壳，载体，涂层，催化剂

Shell, support, coating, catalysts (rare metal, like Pb, Pt, Rh)

Platinum-Palladium-Rhodium alloys

车用催化剂的典型结构

1—堇青石陶瓷蜂窝载体 2— $\gamma\text{-Al}_2\text{O}_3$ 活性涂层 3—催化活性物质

Key rules when using TWC

(三效催化转换装置的使用原则)

➤ Mixture c

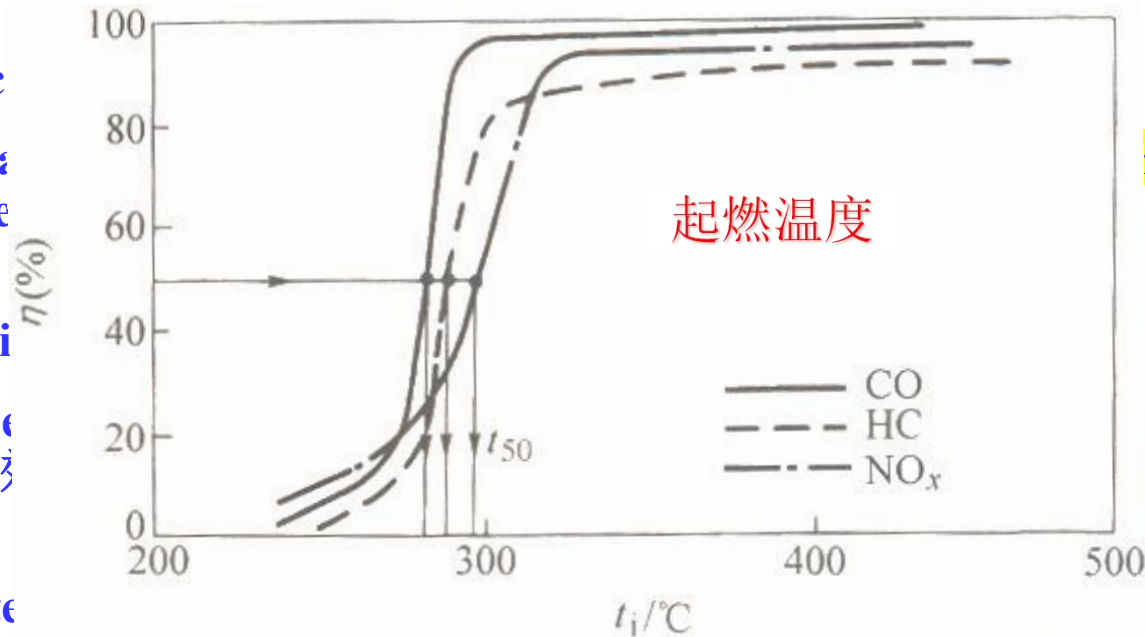
Air-fuel ratio
If mixture

➤ Fuel quality

cannot use
catalyst effectively

➤ Exhaust temperature

Catalytic converter works when its temperature is **over 350°C**, →so it is installed in the back of Exhaust manifold, **close to the exhaust port**.



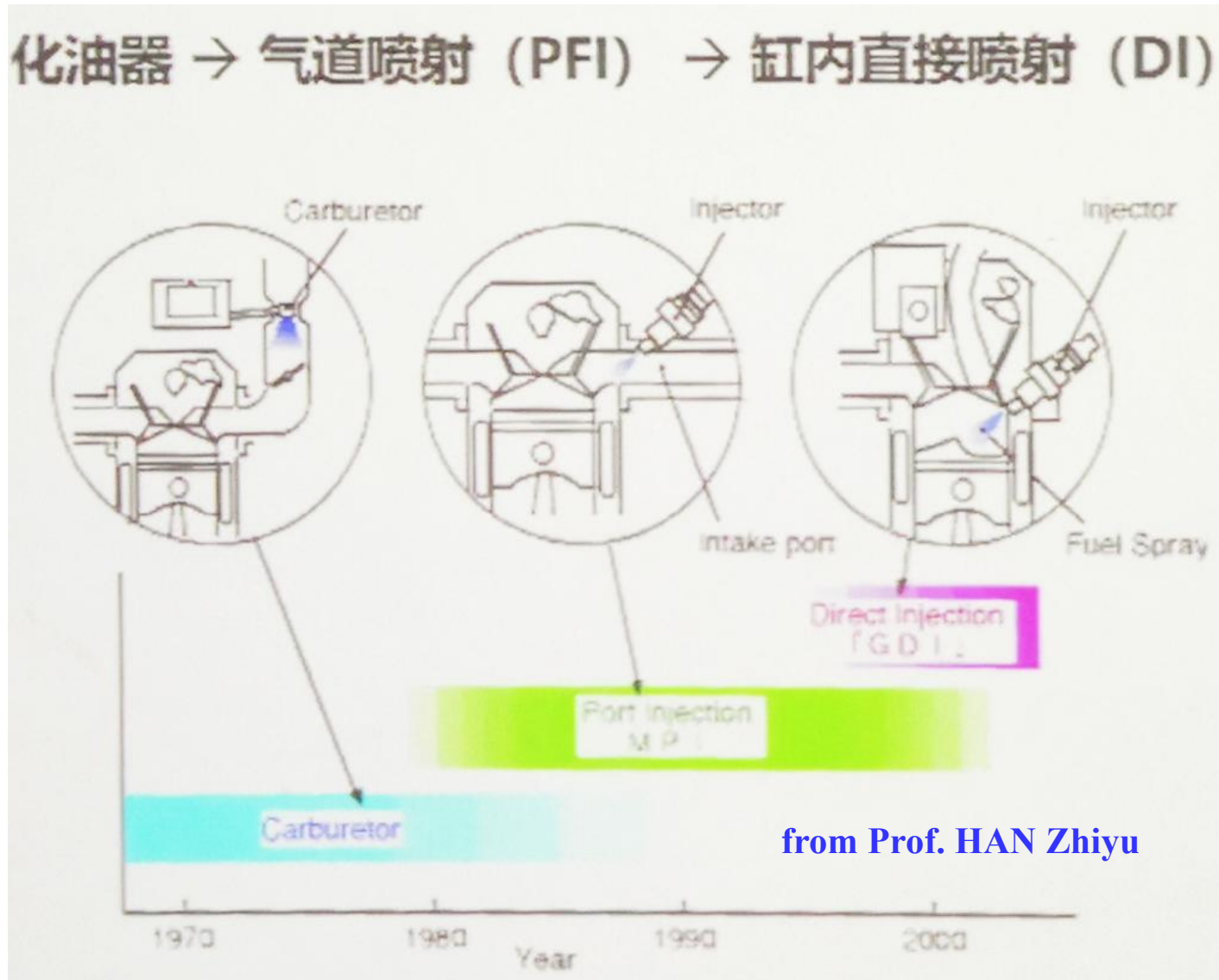
different fuel).
dramatically.

ible catalyst (使

Heat Management is required in control strategies.

5 Gasoline particulate filter (GPF)

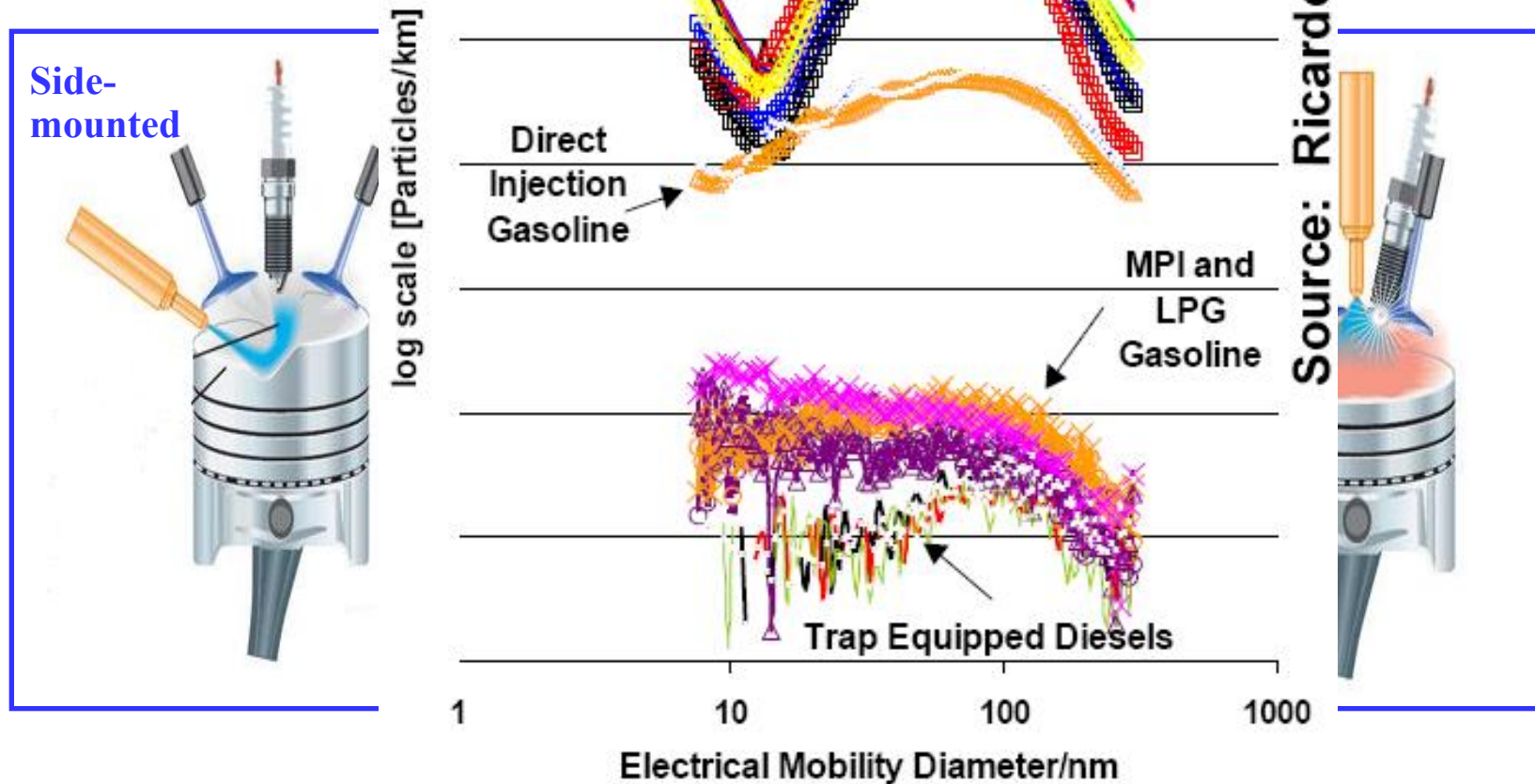
Gasoline
engine
revolution



from Prof. HAN Zhiyu

Comparison of Particle Emissions from SMPS.7: All Vehicles and Fuels - 50kph

GPF is almost same as DPF for diesel engines.

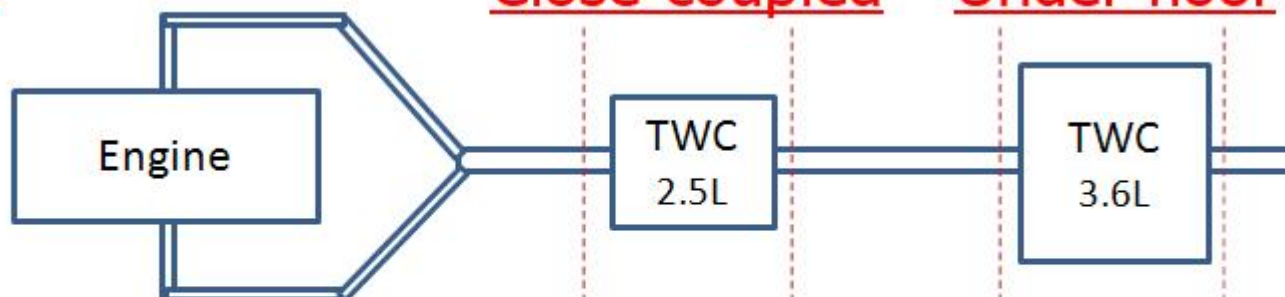


System

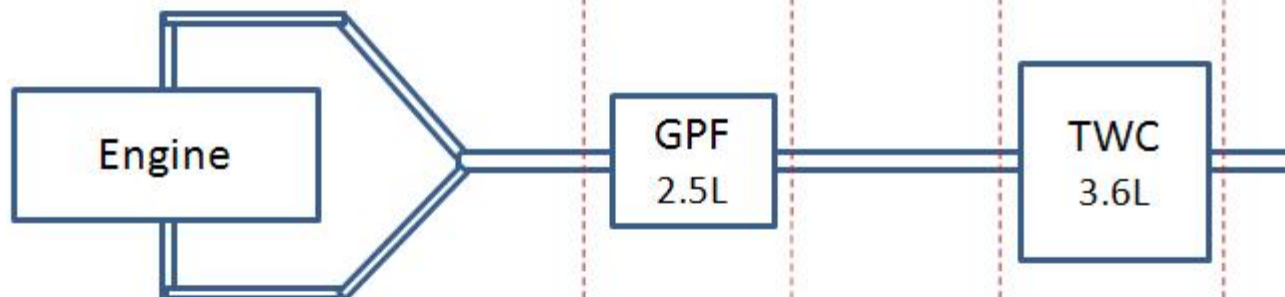
Close-coupled

Under-floor

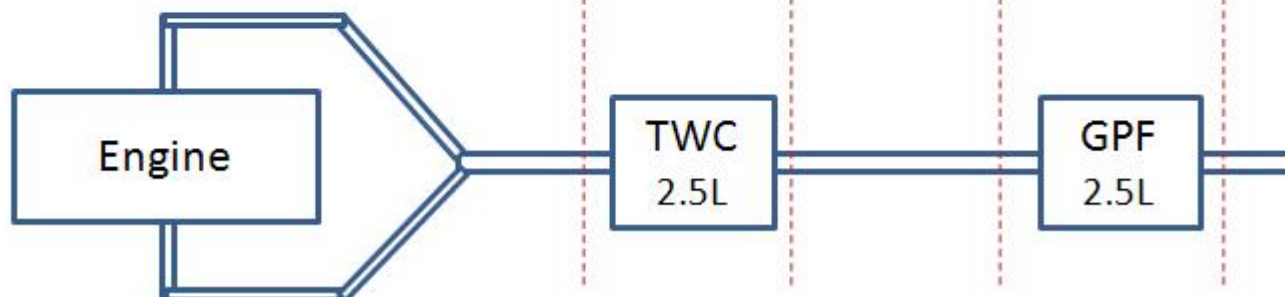
A



B



C



§ 11.3 Emissions Control Devices in Diesel Engine (柴油机的排放控制装置)

1. Characteristics of Diesel engine emissions
2. EGR system
3. Diesel Particulate Filter (DPF)颗粒捕捉器
4. NO_x reduction catalytic converter (NO_x还原催化转化器)
5. Diesel oxidation catalytic converter (柴油氧化催化转化器)

1. Emissions Characteristics of Diesel vs Gasoline engines

(1) Weaker fuel evaporative emissions because of its higher boiling temperature.

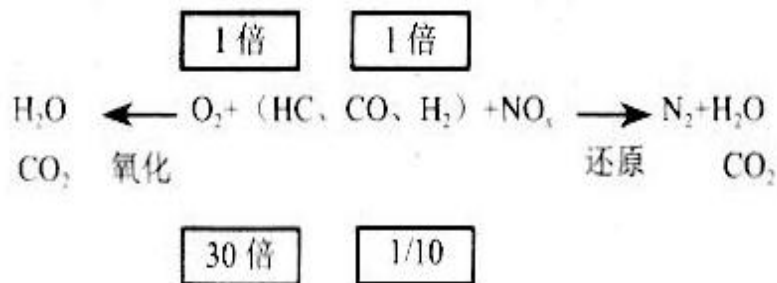
(2) HC and CO emissions are much less, and the **NO_x** emission is in the same order of magnitude(同一数量级).

(3) Particulate matter (**PM**) emission is high.

柴油机与汽油机主要排气成分对比

类别	排气成分及含量/%				
	NO _x	O ₂	HC	CO	H ₂
汽油车排气	0.05~0.15	0.2~0.5	0.03~0.08	0.3~1.0	0.1~0.3
柴油车排气	0.04~0.08	6~15	0.01~0.05	0.01~0.08	0.01~0.05

NO_x 还原体系的概念



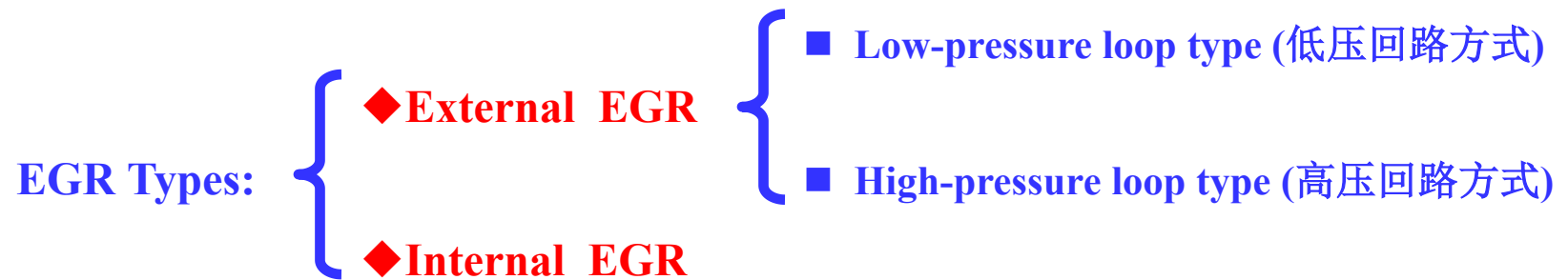
Difficulties:

(1) Exhaust is always in **oxidation atmosphere (excess air coefficient > 1)**, the **NO_x** reduction is hard; 废气总是处于氧化氛围（过量空气系数大于1）中，NO_x难以还原；

(2) Exhaust temperature is lower than that of gasoline engine, soot oxidation is hard; 排气温度明显低于汽油机的排气温度，碳烟难以氧化；

(3) Exhaust contains a lot of **SO_x** and particles, cause catalyst poisoning easily. 排气中含有大量的SO_x和微粒，容易导致催化剂中毒。

2. EGR system



IN-Cylinder Control Method

(1) External EGR

■ Low-pressure loop type(低压回路方式)

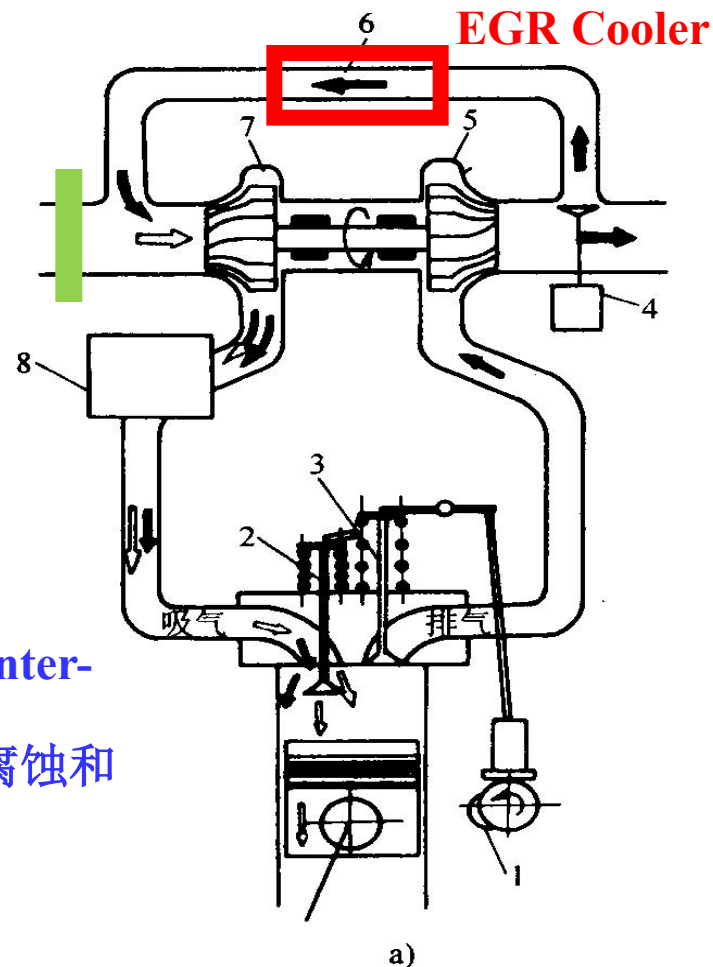
Connect compressor inlet side(压气机7入口端) with the exhaust turbine outlet 5 (废气涡轮5出口) to realize the EGR directly.

Due to compressor inlet side 7 is negative pressure(负压), exhaust turbine outlet 5 is positive pressure(正压), so it can drive exhaust to flow in pipe 6 easily.

But the exhaust flows through the compressor and inter-cooler 8(中冷器8), it's easy to cause corrosion in the compressor and pollution in inter-cooler(压气机的腐蚀和中冷器的污染).

EGR rate is determined by the pressure difference between the exhaust and intake.

Add a throttle in the intake pipe of a diesel engine.



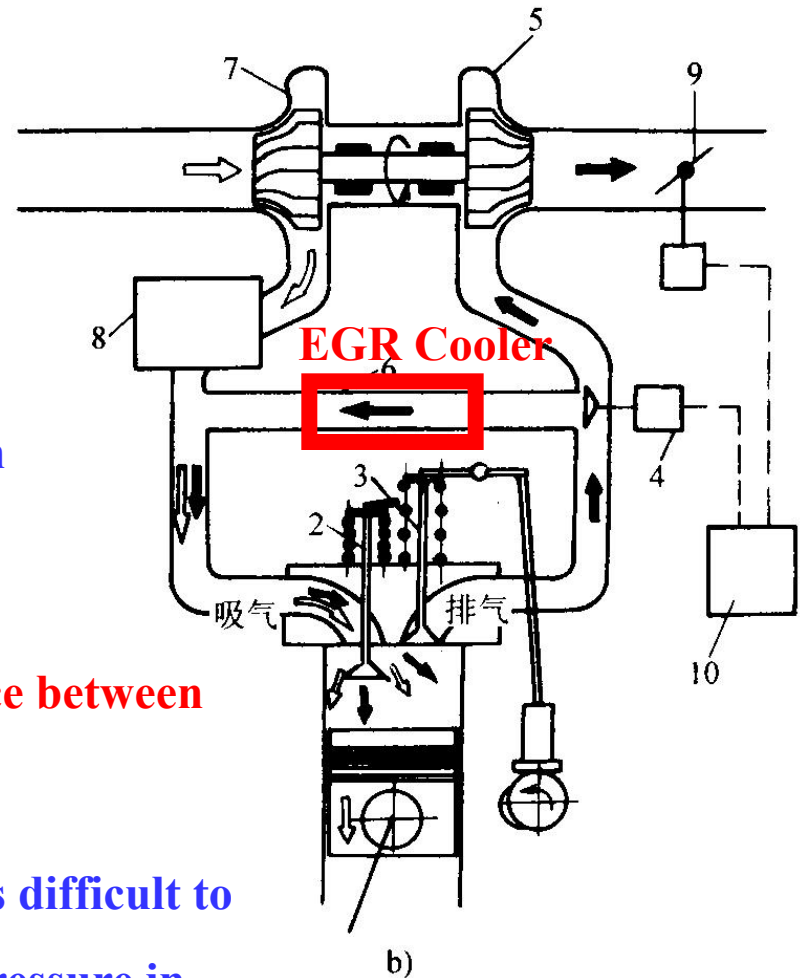
■ High-pressure loop type(高压回路方式)

Connect intake manifold after compressor outlet with exhaust manifold before turbine inlet to realize the EGR directly.

Because the exhausts do not flow through the compressor and inter-cooler, there is no corrosion and pollution in them.

EGR rate is determined by the pressure difference between the exhaust and intake.

In the medium and heavy loads(中、大负荷), it's difficult to realize high EGR rate because of the increased pressure in intake manifold.



(2) Internal EGR

◆ Reduce the valve overlap duration, even to be negative overlap $\alpha + \beta$ -- by VVA

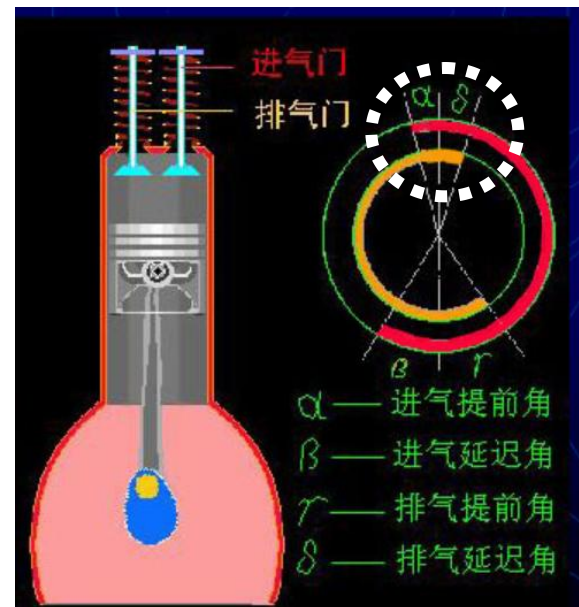
◆ Use pulsation of gas in inlet and exhaust pipes(利用进、排气管中的气体脉动进行EGR的方式). During the intake stroke, make one cylinder appear positive pressure wave near the exhaust valve. At this point, open the exhaust valve, it can realize the EGR.(使某一缸在进气过程中, 其排气门处出现正压波, 此时, 如果能再次开启排气门, 就可实现EGR)。

◆ Add internal EGR CAM(增设EGR专用凸轮). At appropriate moment in intake stroke, open the exhaust valve to make exhaust flow back into the cylinder with the help of EGR CAM(使排出的废气回流到气缸内部), to realize the EGR.

Advantages:

need no exhaust throttle (排气节流), → not affect the pumping losses (泵气损失) → not affect fuel economy;

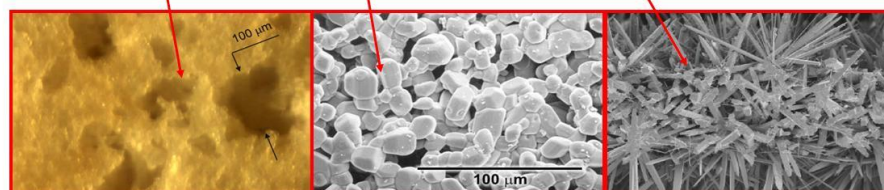
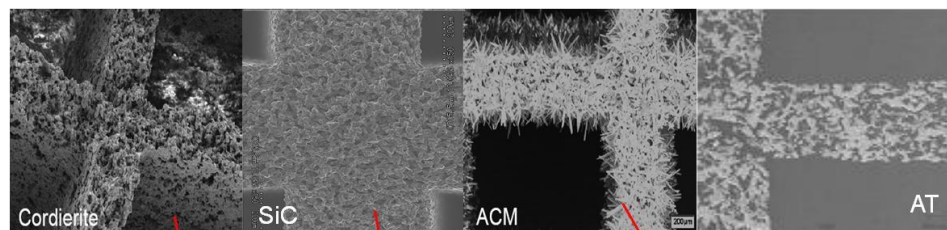
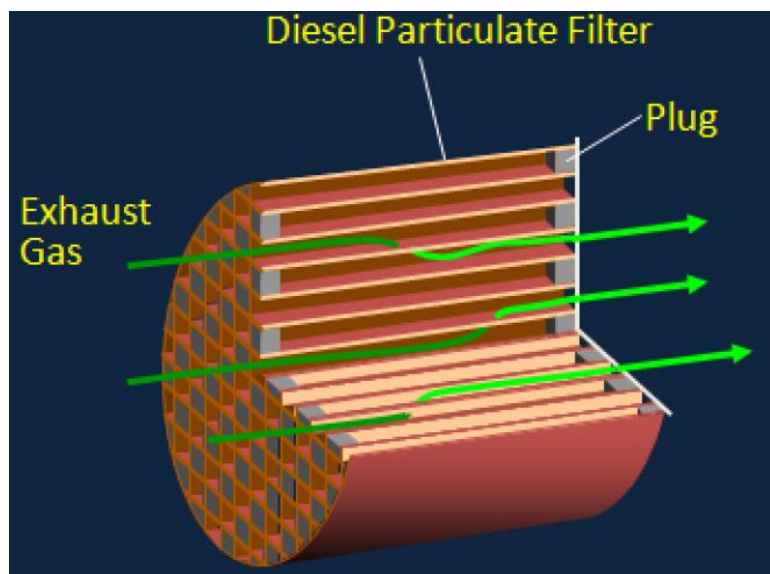
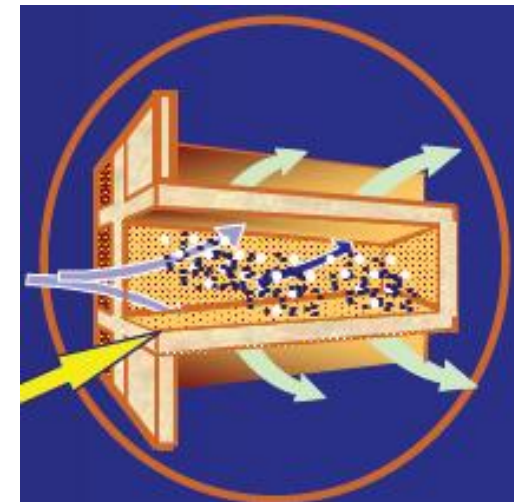
No EGR valve and EGR pipe → structure is simple.



3. Diesel Particulate Filter (DPF)

柴油机颗粒过滤器

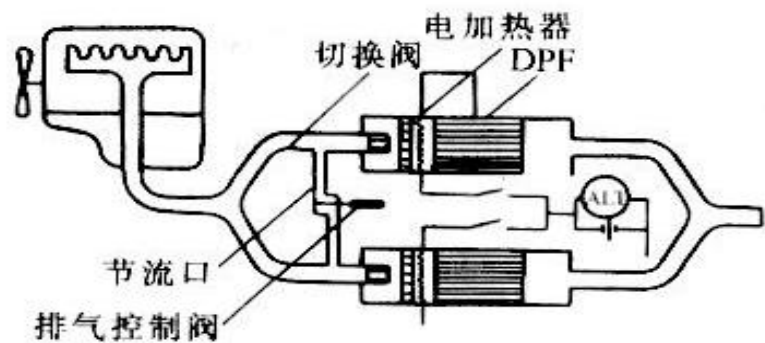
- Requirement** {
- (1) high filtration efficiency 高过滤效率
 - (2) low exhaust resistance 低排气阻力
 - (3) high temperature resistant 耐高温



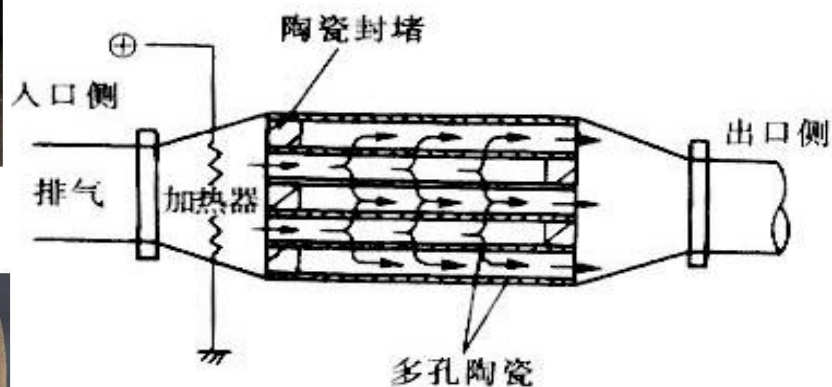
AT: Aluminum Titanate (Corning)
ACM: Advanced Ceramic Material (Dow)

DPF regeneration method:

- (1) electric heating regeneration 电加热再生
- (2) burner heating regeneration 燃烧器加热再生
- (3) counter blowing regeneration 反吹式再生
- (4) continuous catalyst regeneration 连续催化再生



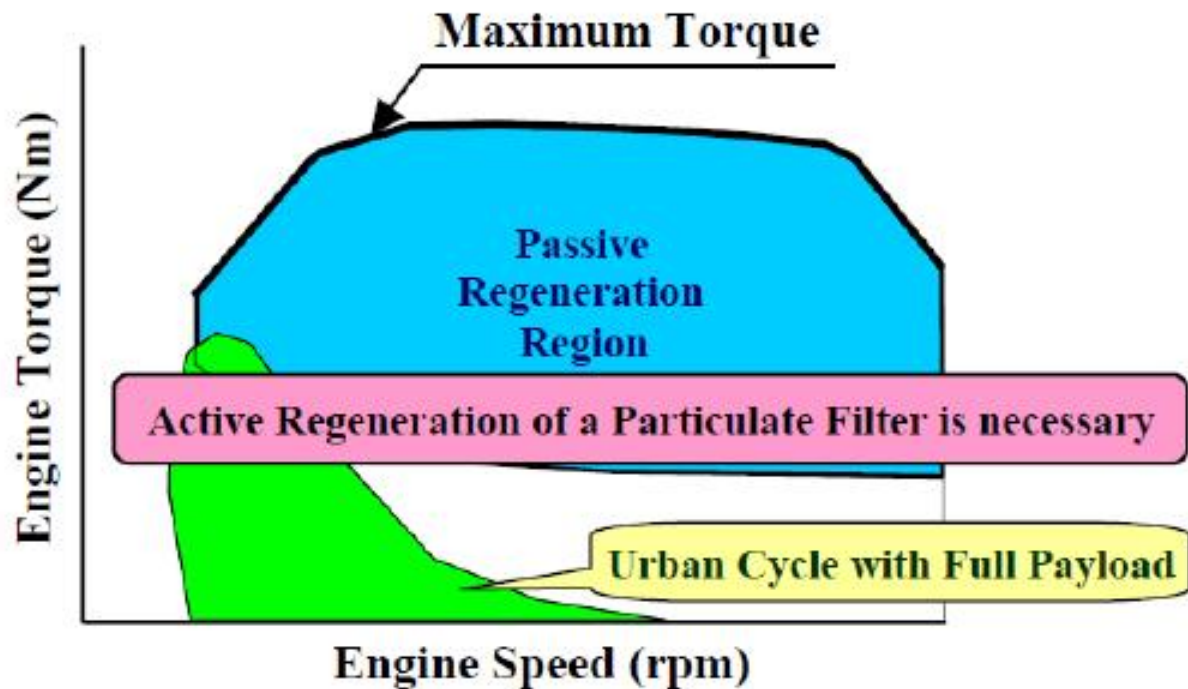
(a) DPF及控制系统



(b) DPF 电加热再生

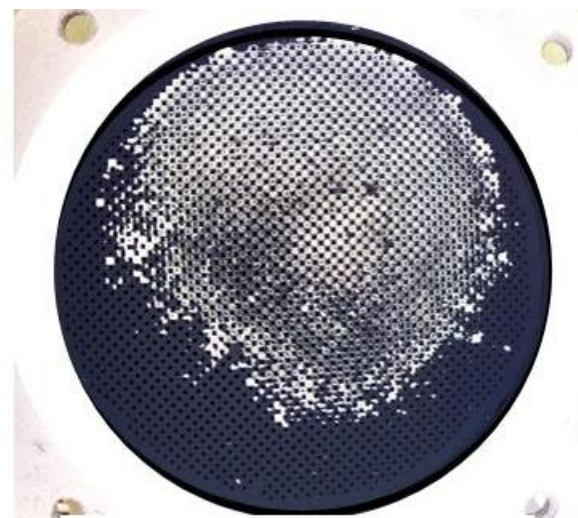
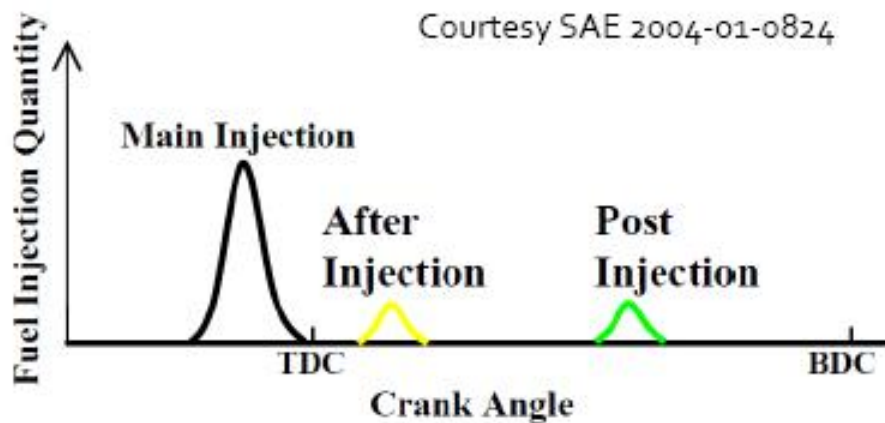
DPF 的电加热再生控制系统



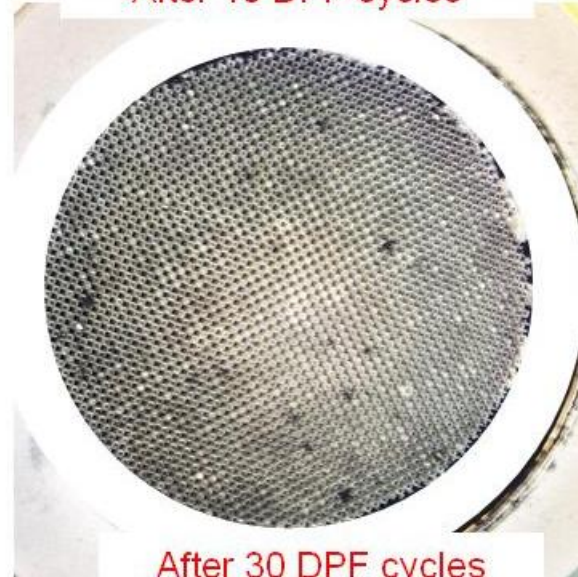


主动、被动
再生相结合

机内、机外
相配合

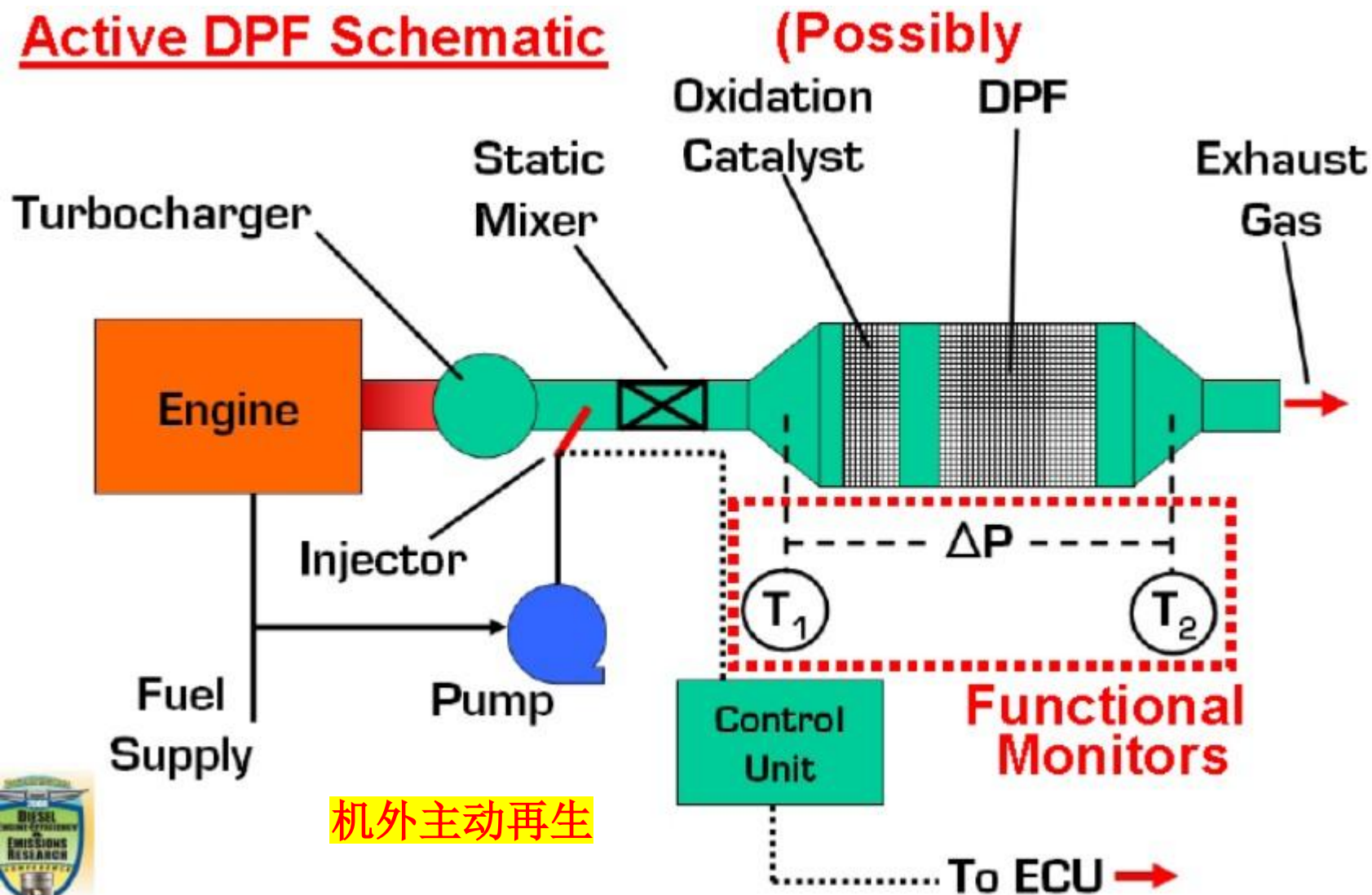


After 10 DPF cycles



After 30 DPF cycles

Active DPF Schematic

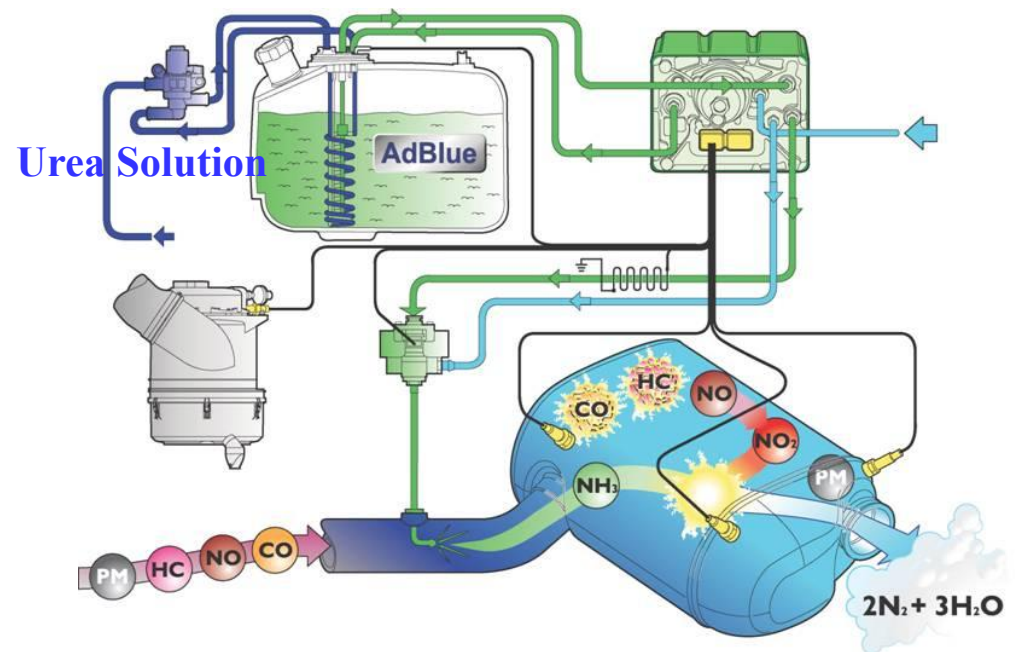


4. NO_x reduction catalytic converter (NO_x还原催化转化器)

Types:

- Selective Catalytic Reduction (SCR) (选择性催化还原);
- Selective Noncatalytic Reduction(**SNCR**)(选择性非催化还原);
- Non Selective Catalytic Reduction(**NSCR**)(非选择性催化还原).

SCR { NH₃ - SCR
HC - SCR



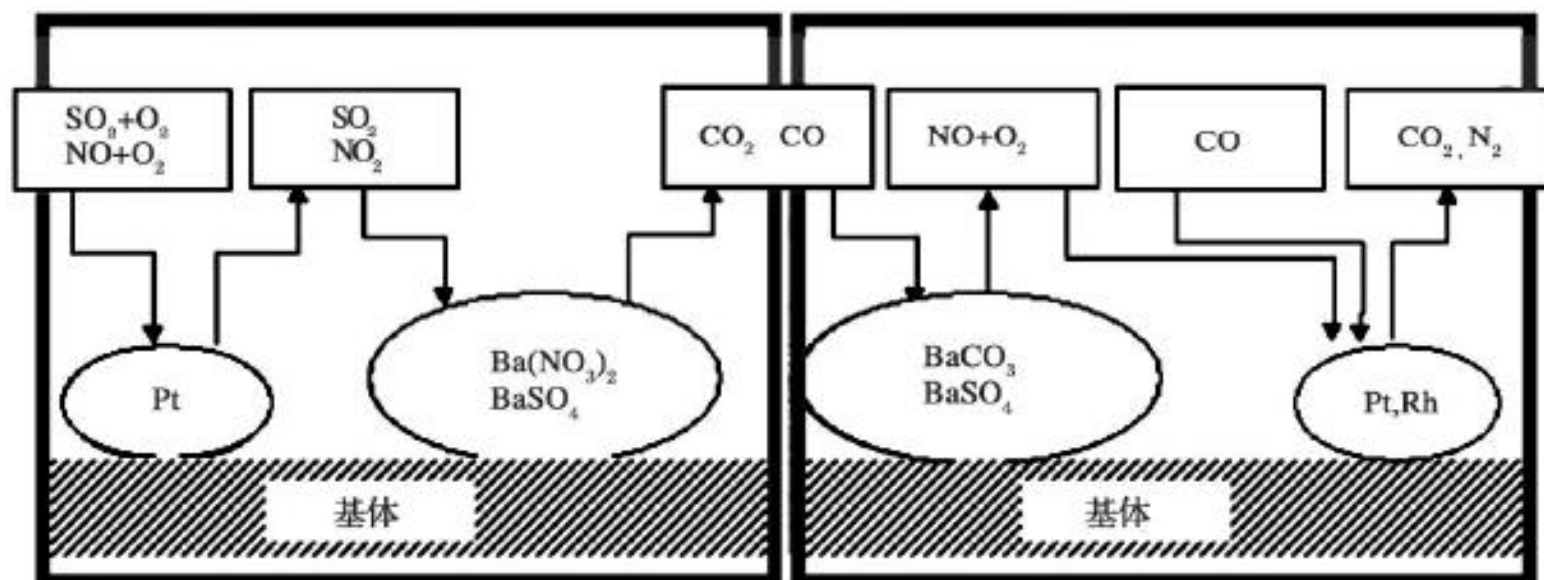
Types:

► Lean NO_x Trap (LNT) (稀薄工况NO_x吸附)

稀燃时，临时存储NO_x。NO通过Pt催化氧化成NO₂，NO₂被碱金属化合物吸附而被捕集；

当吸附达到饱和后，再生（富油燃烧条件），解析NO_x，铈、铂金催化，与HC、CO、H₂反应还原为N₂。

对超低S含量燃料，NO_x降低效率可达90%。



稀燃 NO_x 储存阶段: 空燃比 $\lambda > 1$

富燃再生阶段: 空燃比 $\lambda < 1$

5. Diesel oxidation catalyst (DOC) (柴油机氧化催化转化器)

Fundamental:

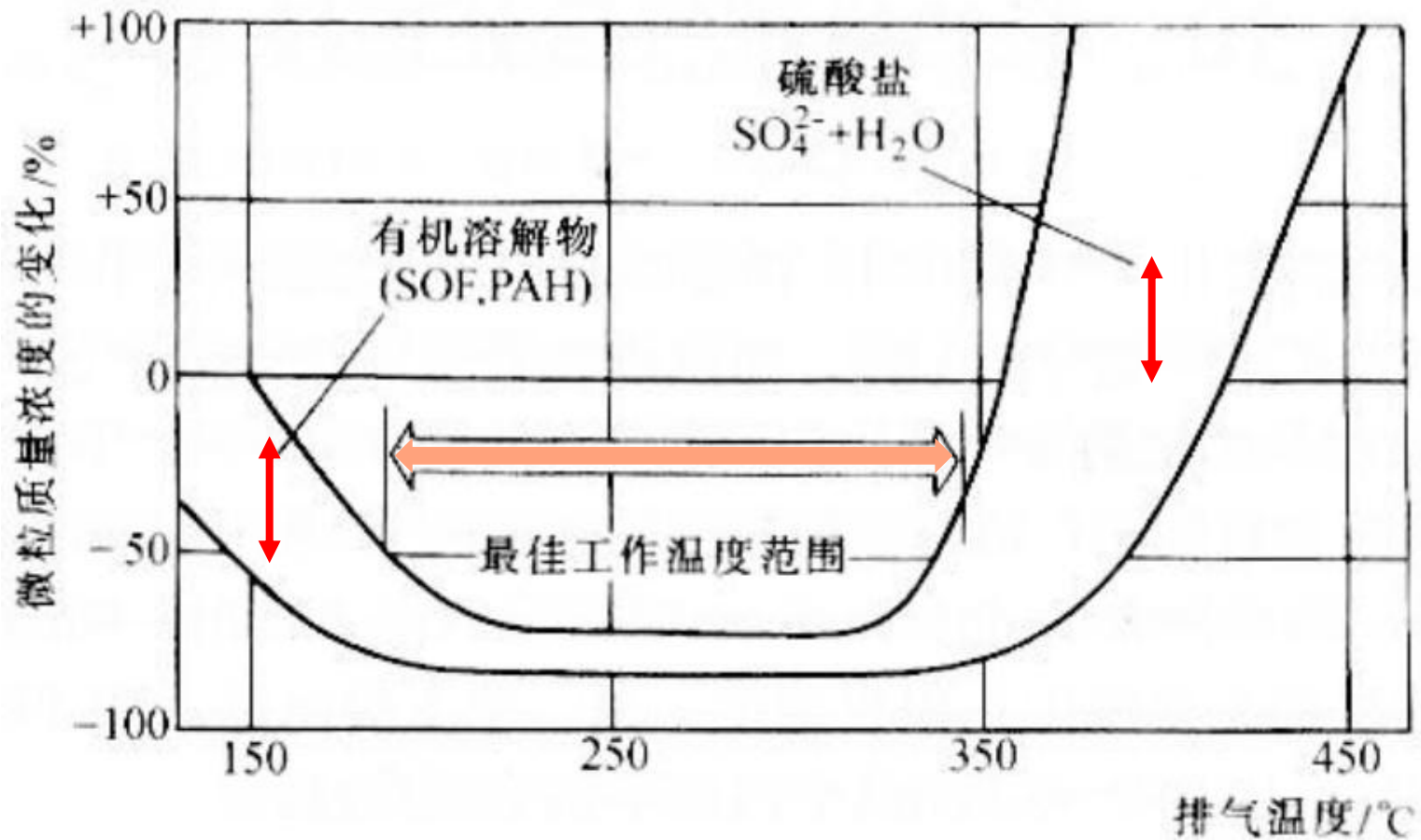
With the help of catalyst **Pt or Pd**,

➤ CO+HC are oxidized to $\text{CO}_2 + \text{H}_2\text{O}$

➤ SOF in PM can be also oxidized into $\text{CO}_2 + \text{H}_2\text{O}$

➤ Polycyclic aromatic hydrocarbons(多环芳香烃), acetaldehyde(乙醛) can be also oxidized.

But sulphur(硫) in diesel can also be oxidized to RSO_4 and RSO_3 which are part of PM, maybe increase PM emission. So the fuel for diesel engines with DOC should **contain very little sulphur**.



Heat Management is required in control strategies.

Typical diesel exhaust aftertreatment system

(典型的柴油机排气后处理方案)

EGR + DOC + DPF + SCR

